



2567

31 901-2002

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Ubuntu Server,

Proxmox VE LXC Container

Linux

CLI	DNS/DHCP, Nginx, MariaDB, Node.js, SSH, Firewall, FTP,
	Clone GitHub

Server	Service, Port, Log, Permission
Client	

$$\left(\begin{array}{c} \text{ } \end{array} \right)$$

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7	/	Git	Clone	GitHub	
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(Network Operating System for
Server)

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1. NOS, OSI 7 Layer, TCP/IP
 2. Ubuntu Server, Proxmox VE, LXC
User, Permission, SSH, Firewall Network Service
 3. DNS, DHCP, Web, Database, File, FTP,
Proxy, AAA, IoT Deploy Web Application
 - 4.
-
1. NOS, OSI/TCP-IP
Virtualization, Container Server
 - 2.
 3. Deploy Web
Application/Server Service

		OSI 7 Layer	TCP/IP
Protocol			LAN
	TIA/EIA	Proxmox Hypervisor	
		LXC Container	
	Firewall	SSH	
		DNS Server, Web Server, Database Server,	
DHCP Server	File and Resource Sharing Server		Proxy
Server, AAA Server, Container Platform		IoT Platform	
	SSL Certificate		
	Nginx Reverse Proxy	Deploy Web Application	
Git, Node.js	MariaDB	DevOps	

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40 106	5	NOS-0 1	NOS, OSI / TCP-IP	LAN	
		NOS-0 2	Ubuntu Server Linux CLI		Checklist
		NOS-0 3	DNS DHCP		
		NOS-0 4	Web / Database / Application Server	Nginx, MariaDB Node.js	Service / Port / Config
		NOS-0 5	User, Permission SSH		SSH Key
		NOS-0 6	Firewall	Nmap	Firewall Audit
		NOS-0 7	FTP	vsftpd	Upload / Download

				FileZilla	
		NOS-08	Proxmox VE LXC	Hypervisor, Network Container	Web GUI / LXC
		NOS-09	Deploy Web Application HTTPS	Nginx / PM2/SSL	Client
		NOS-10	Git Clone GitHub	Public Repository Server	Git Clone Checklist
		NOS-11	Clone	Dependencies	Deployment Checklist

(Job)				
(Duty)	(Task)	()		
	1.1 NOS, Client-Server	NOS-01	Server/Client/Switch/Router	
	1.2 OSI/TCP-IP Protocol		Layer, Encapsulation, Port	
	1.3 LAN		UTP, T568A/B, RJ-45	LAN Tester
	2.1 Ubuntu Server	NOS-02/ NOS-03	Installation, Directory, User, Network	
	2.2 Linux CLI DNS/DHCP		Command, DORA, DNS Record	Name/IP
3. Server Stack	3.1 Git Nginx	NOS-04	Package, Document Root, systemd	
	3.2 MariaDB, Node.js Reverse Proxy		SQL, Runtime, proxy_pass	Stack Log
(Job)				

(Duty)	(Task)	()		
	4.1 User, Permission SSH Key	NOS-06	Least Privilege, chmod/chown, sshd	Hardening
	4.2 Port UFW/Nmap		TCP/UDP, Bind, Rule, Scan	
	5.1 vsftpd Passive Port	NOS-07	Control/Data Connection, Config	Firewall
	5.2 FileZilla		Message Log, Listing, Transfer	Upload/Download Error
6. Proxmox/LXC/Deployment	6.1 Proxmox VE LXC	NOS-09	Hypervisor, Bridge, Storage, Template	Provision Container Network
	6.2 Deploy Web Application HTTPS		Build, Nginx, PM2, TLS	Restart
(Job)				
(Duty)	(Task)	()		
	7.1 Git Clone Public	NOS-11	Git, HTTPS URL, Repository,	Container

	Repository		Clone	
	7.2 Dependencies		README, package.json, npm, PM2/Nginx	Process/Port

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: / 35 20
15 20 100

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1	L N	2	3	5
2	Ubuntu Server Linux DNS/DHCP	2	8	10
3	Web Server, Database Node.js Server Stack	3	12	15
4	SSH Server	2	8	10
5	FTP Server	2	8	10
6	Proxmox VE, LXC Container	2	13	15
7	Git Clone GitHub Server	2	8	10
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L N						
2. Ubuntu Server Linux DNS/DHCP	K2	S3	2	p2	2/8	10
3. Web Server, Database Node.js Server Stack	K3	S3	3	p3	3/12	16
4. SSH Server	K3	S3	3	p3	2/8	13
5. FTP	K3	S3	3	p3	2/8	13

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Server						
6. Proxmox VE, LXC Container	K3	S3	3	p3	2/13	18
7. Clone GitHub Server	Git K3	S3	3	p3	2/8	23
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Log Ubuntu Server, Proxmox VE LXC SSH Firewall Port Service						
.....						
K1 = K2 = K3 = K4 = K5 =	S1 = S2 = S3 = S4 = S5 =			1 = 2 = 3 = 4 =		

					/	
K6 = _____	_____				5 = _____	
1						
p1 = p2 = p3 = p4 = p5 = _____						

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	LAN	2 . 3
/ NOS, OSI/TCP-IP, RJ-45		.

1.

OSI/TCP-IP

UTP T568A/T568B

LAN Tester

2. /

2.1 : 40 106

5

1) :

Client, Protocol OSI Server,
LAN 1-8 70

2) :

3) LAN
(Performance Evidence):

1 LAN

4) (Knowledge Evidence):

OSI 7 Layer

T568A/T568B

2.2 :

Server

3.

3.1 Network Operating
System

3.2	Server, Client, Switch, Router, Gateway
Access Point	
3.3	OSI 7 Layer TCP/IP Protocol
3.4	Encapsulation De-encapsulation
3.5	UTP
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4.3	OSI 7 Layer Protocol
4.4	Client Server
4.5	T568A T568B
4.6	RJ-45 LAN Tester
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	Network Operating System,
Client-Server,	, OSI 7 Layer, TCP/IP, Protocol
Port,	Encapsulation, UTP,
T568A/T568B, Straight-Through/Crossover	
RJ-45	
6.	
6.1	Server, Client
6.2	NOS Animation
6.3	OSI 7 Layer Protocol, Port Layer
6.4	RJ-45
	T568B
6.5	1-8
LAN Tester	

6.6

Server

7.

7.1

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NOS,

OSI Model

7.2

UTP,

RJ-45,

,

LAN

Tester

7.3

1

1

8.

8.1

OSI 7 Layer

T568A/T568B

8.2

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LAN

9.

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70

Layer

LAN

9.2

9.3

LAN

Tester

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10.1

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2)

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/ NOS, OSI/TCP-IP, RJ-45		.

1.

LAN Tester UTP T568A/T568B OSI/TCP-IP

2. /

2.1 : 40 106

5

1) : Server,
70

Client, Protocol OSI 1-8

LAN

2) :

3) LAN
(Performance Evidence):

1 LAN

4) (Knowledge Evidence):

OSI 7 Layer T568A/T568B

2.2 :

Server

3.

3.1 Network Operating
System

3.2	Server, Client, Switch, Router, Gateway		
Access Point			
3.3	OSI 7 Layer	TCP/IP Protocol	
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3.5	UTP		
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4.2			
4.3	OSI 7 Layer	Protocol	
4.4		Client	Server
4.5		T568A	T568B
4.6	RJ-45	LAN Tester	
4.7			
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	Web, Database, File Server	DNS Server	Client
5.2			
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, Gateway		Access Point	
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	Encapsulation		

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UTP	4		
T568A T568B		1-8	
Straight-Through LAN Tester			

6. /

6.1		
Physical Layer		
6.2	Switch, Router, Gateway	Access Point
6.3	T568B	
LAN Tester		

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2567; Cisco Networking Academy. Networking Basics;

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8. ()

8.1	:	Layer
Layer		
8.2	: Switch	LAN, Router
Gateway, AP		
8.3		
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	Ubuntu Server Linux DNS/DHCP	2 . 8
/ Ubuntu Server, Linux CLI, DNS DHCP Fundamentals		.

1.

	Ubuntu Server	Linux
	DNS DHCP	
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1)	:	80
	DHCP DORA	DNS Record
2)	:	Checklist
	Terminal	
DNS/DHCP		
3)	(Performance Evidence):	Ubuntu
Server	CLI	
DHCP/DNS		
4)	(Knowledge Evidence):	
Directory	DORA	DNS
Record		
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3.1	Ubuntu Server	
3.2	Directory	Linux
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- 3.4 IP Address, Route, DNS Service
- 3.5 DHCP DORA DNS Record

4.

- 4.1 GUI CLI

Server

- 4.2 Ubuntu Server

- 4.3 pwd, ls, cd, mkdir, touch, cp, mv, rm nano

- 4.4 apt systemctl

- 4.5 ip a, ip route, ping

4.6 DHCP DOR , CN ME, MX, TXT Record

- 4.7

5.

Linux, GUI/CLI, Root ,
 Directory, Absolute/Relative Path,
 , systemd, , DHCP
 DORA, Scope, Lease, Reservation, DNS Recursive/Authoritative
 DNS Record

- 6.

- 6.1 Server

Linux CLI

- 6.2 Ubuntu Server

Disk, User, Hostname Network

- 6.3 Ubuntu Server

Directory

- 6.4

nano

6.5	/ nimation	DHCP
DOR		DNS
6.6	IP, Route	DNS
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7.1	2-3	Ubuntu Server, Linux CLI,
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7.2	Ubuntu Server ISO,	Server/VM/LXC, Terminal
	Network Diagram	
7.3	2	Linux
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DORA	DNS Record	
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	Installation Checklist, Command Checklist,	
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	Ubuntu Server Linux DNS/DHCP	2 . 8 .
/ Ubuntu Server, Linux CLI, DNS DHCP Fundamentals		

1.

- Ubuntu Server Linux
- DNS DHCP
2. /
- 2.1 : 40 106
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- 1) :
- 80
- DHCP DORA DNS Record
- 2) : Checklist
- Terminal
- DNS/DHCP
- 3) (Performance Evidence): Ubuntu
Server CLI
- DHCP/DNS
- 4) (Knowledge Evidence):
Directory DORA
- DNS Record
- 2.2 :

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- 3.1 Ubuntu Server
- 3.2 Directory Linux

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3.4	IP Address, Route, DNS		Service
3.5	DHCP DORA		DNS Record
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4.1		GUI	CLI
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4.2	Ubuntu Server		
4.3	pwd, ls, cd, mkdir, touch, cp, mv, rm		nano
4.4	apt		systemctl
4.5	ip a, ip route, ping		
4.6	DHCP DORA		A, CNAME, MX, TXT Record
4.7			
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Linux			Server
	ISO		Network Storage
	Hostname		
		IP	
5.2		Linux	
	Linux		command option argument
ls -la /etc			pwd, ls, cd, mkdir, touch, cp, mv, rm
nano		/	Absolute Path
Relative Path			sudo
5.3			
	sudo apt update		
	apt install		
	systemctl start, stop, restart, enable		status
	ip a, ip route, ping		hostname -I
5.4	DHCP		DNS

DHCP	IP, Subnet Mask, Gateway	DNS	
Discover, Offer, Request, Acknowledge		DORA	DNS
	IP Recursive DNS		
Authoritative DNS		Record	A,
CNAME, MX	TXT		IP Route
DNS			
6.	/		
6.1	/home/student/lab2		note.txt
6.2	DHCP DORA		
Client			

6.3 ping IP

7.
 Ubuntu Server Documentation; man pages Linux; ISC
 DHCP Documentation; Cloudflare Learning Center: DNS;
 2-3

8. ()

```
8.1      : mkdir -p /home/student/lab2, cd
/home/student/lab2, touch note.txt, pwd      ls -l
```

8.2 Discover, Offer, Request, Acknowledge, Lease

8.3	DNS Server	Name Resolution
	resolvectl, nslookup	dig
	/etc/resolv.conf	

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	Server, Database Node.js Server Stack	3 . 12
/ Git, Nginx, MariaDB, Node.js Nginx Reverse Proxy		.

1.

Git, Nginx, MariaDB Node.js
Ubuntu Server
Nginx Reverse Proxy

2.

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2.1 : 40 106
5
1) :
Restart
Reverse Proxy
Node.js
2) : Service, Port, Config
Client
3) (Performance Evidence):
Nginx / Node.js Application
Nginx Site Configuration
4) (Knowledge Evidence): Server
Stack Request
Client Application/Database
2.2 : Web/Database Server
Deploy Web Application DevOps

3.

3.1 Nginx Web Server
3.2 MariaDB
3.3 Node.js HTTP Application
3.4 Git Source Code Server

3.5	Nginx Reverse Proxy	Log
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4.1	Nginx, MariaDB, Node.js	Git
Server Stack		
4.2	Nginx	Document Root
4.3	systemctl	nginx -t
4.4	Database, Table	User
4.5	Node.js HTTP Server	
4.6	proxy_pass	Client
4.7		Deploy
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	Server Stack,	
	Git/Nginx/MariaDB/Node.js, Document Root	Virtual Host, SQL
		, Node.js HTTP Server, npm,
	Nginx Reverse Proxy, systemctl, Port	Log
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6.1		Browser
Nginx	Node.js	MariaDB
6.2		Git Nginx
	HTML	Server
6.3		Database, User Grant
6.4		Node.js HTTP Server
	ss -tulpn	
6.5	Nginx Reverse Proxy	
	Site Configuration	nginx -t
Reload		
6.6	Client	Access/Error Log Stack
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7.1	Nginx, MariaDB, Node.js, Git	Reverse
Proxy		
7.2	Ubuntu Server/LXC, Web Browser, Terminal	Source
Code		
7.3	3	Request
Flow	Server Stack	

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8.1	Server Stack
	Request Client Application/Database
8.2	
	Nginx / Node.js
Application	Nginx Site Configuration

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9.1	Config
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9.2	Service/Port/Config Request Client
9.3	Server Stack Checklist, Config,

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- 10.2
- 10.3
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/ Git, Nginx, MariaDB, Node.js Nginx Reverse Proxy		.

1.

Ubuntu Server Git, Nginx, MariaDB Node.js
Nginx Reverse Proxy

2.

2.1 : 40 106
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1) :
Restart

Reverse Proxy
Node.js
2) : Service, Port, Config
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3) (Performance Evidence):
Nginx / Node.js Application
Nginx Site Configuration
4) (Knowledge Evidence): Server
Stack Request
Client Application/Database
2.2 : Web/Database
Server Deploy Web Application DevOps

3.

3.1 Nginx Web Server
3.2 MariaDB

3.3 Node.js HTTP Application
 3.4 Git Source Code Server
 3.5 Nginx Reverse Proxy Log

4.

4.1 Nginx, MariaDB, Node.js Git
 Server Stack
 4.2 Nginx Document Root
 4.3 systemctl nginx -t
 4.4 Database, Table User
 4.5 Node.js HTTP Server
 4.6 proxy_pass Client
 4.7 Deploy

5.

5.1 Server Stack Request
 Nginx HTTP Request 80/443 Static File
 Application Reverse Proxy, Node.js
 MariaDB Git
 Source Code

5.2 Nginx Web Server

apt install nginx systemctl status nginx
 /var/www/html Site
 Configuration /etc/nginx/sites-available
 sites-enabled Config nginx -t
 Reload Syntax Error

5.3 MariaDB Node.js
 MariaDB root
 GRANT Node.js JavaScript
 Server npm HTTP Server
 3000 Process, Port Error Message

Application

5.4 Git Nginx Reverse Proxy
 Git clone Server git pull
 Reverse Proxy location / proxy_pass
 http://127.0.0.1:3000

Nginx
 systemctl status, ss -tulpn, curl Log nginx -t,
 6. /
 6.1 Request
 Nginx Node.js MariaDB
 6.2 Nginx 502 Bad
 Gateway
 6.3 Reverse Proxy
 systemctl, ss curl
 7.

Nginx Documentation; MariaDB Server Documentation;
Node.js Documentation; Git Documentation; Server
Stack Reverse Proxy

8. ()

8.1 Browser -> Nginx -> Node.js -> MariaDB

8.2 Application Process/Port, curl 127.0.0.1:3000, proxy_pass,
 nginx -t Error Log

8.3 Checklist

Config

Client

4.

- 4.1
- 4.2 Read/Write/Execute
- 4.3 Owner/Group ls -l
- 4.4 SSH Key Public/Private Key
- 4.5 TCP/UDP, Port Bind 127.0.0.1/0.0.0.0
- 4.6 ss, curl, ping, nmap UFW

4.7 Rule

5.

User/Group/sudo, Least Privilege, Linux Permission, chmod/chown, SSH Key, sshd_config, TCP/UDP, Port Bind address, ip/ss/ping/curl/nmap, UFW Default Policy, Allow/Deny IP Whitelisting

6.

- 6.1 root
- 6.2 User/Group Owner Permission
- 6.3 SSH Key Private/Public Key
- 6.4 IP, Route, Service Port ip a, ip route, systemctl ss -tulpn
- 6.5 UFW Default Deny Allow Rule

6.6

Nmap

/

Firewall

7.

- 7.1 6-7 User, Permission, SSH, Network Diagnostics UFW
- 7.2 Ubuntu Server/LXC LAN Terminal

7.3	4	Permission/SSH
Firewall Audit		
8.		
8.1	chmod	Least Privilege
	Network Service	
8.2		SSH Key
Configuration UFW Rule	Port Scan	
9.		
9.1		
Permission	SSH Key	UFW
	Nmap	Rule
9.2		
		Login/Access
	Port Scan	
9.3		
Permission Checklist, SSH Hardening Checklist, UFW/Nmap		
Audit Sheet		
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10.1		
10.2		
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2)		

4.

- 4.1
- 4.2 Read/Write/Execute
- 4.3 Owner/Group ls -l
- 4.4 SSH Key Public/Private Key
- 4.5 TCP/UDP, Port Bind 127.0.0.1/0.0.0.0
- 4.6 ss, curl, ping, nmap UFW
- 4.7 Rule

5.

- 5.1 User, Group, sudo Least Privilege
 - adduser
 - usermod -aG sudo
 - root
- 5.2 Linux Permission, chmod chown
 - Permission Owner, Group, Others r=4, w=2, x=1
 - 750 Owner Group /
 - Others chmod Permission chown
 - Owner Group ls -l
- 5.3 SSH Key SSH Hardening
 - SSH Key Private Key Client Public Key
 - ~/.ssh/authorized_keys Server Key
 - PasswordAuthentication PermitRootLogin
 - sshd_config Session
- 5.4 Port Monitoring, UFW Nmap
 - Port TCP/UDP Bind 127.0.0.1
 - 0.0.0.0 Interface ss -tulpn Process/Port,
 - curl , UFW Default Deny
 - Nmap
- 6. /
- 6.1 Permission Owner / / Group
- / Others

6.2	SSH Key	Password
Login		

```

6.3      UFW Rule      SSH      Web      Nmap
        Client

```

7.
 Ubuntu Server Guide: Users and SSH; OpenSSH Manual; UFW
 Manual; Nmap Reference Guide;
 6-7

```

8.      ( )
8.1    chmod 750      Owner=7, Group=5, Others=0
8.2    Key      Permission      Password
      Server

```

8.3 Nmap

		5
	31 901-2002	9-10
	FTP Server	2 . 8
Firewall	/ vsftpd, User Permission, Passive Ports, FileZilla	.

1.

Control/Data Connection Passive Mode
vsftpd
Firewall Upload/Download FileZilla

2.

2.1 : 40 106
5
1) :
Server
Owner/Permission Firewall
2) : Config Service FileZilla
Client Message Log
Login/Listing/Transfer Error
3) (Performance Evidence):
vsftpd.conf FTP upload UFW
Transfer Successful FileZilla
4) (Knowledge Evidence):
Control/Data Connection Passive Port

2.2 : File
Transfer Service, Permission,
Port Firewall

3.

3.1 FTP Control Data Connection
3.2 vsftpd
3.3 FTP User /

3.4	Passive Port Range	UFW
3.5	FileZilla	Message Log
4.		
4.1	FTP	Connection
4.2	Passive Port	Firewall
4.3	vsftpd	
4.4	upload	Owner
4.5	local_enable, write_enable	
chroot_local_user		
4.6	FileZilla	
4.7	Message Log	
5.		
	FTP, Port 21, Control/Data Connection,	
	ctive/Passive Mode, Passive Port Range,	
	vsftpd, Local User, chroot, Directory/Owner/Permission,	
	UFW	FileZilla Site Manager, Transfer Queue
	Message Log	
6.		
6.1		Animation
Login	Port 21	Passive Port
6.2	IP	Container
vsftpd	Service/Port	
6.3	vsftpd.conf	
6.4	ftpstudent	ftp/upload
welcome.txt	Owner/Permission	ls -l
6.5	pasv_min_port/pasv_max_port	UFW
Port 21	Passive Port	Reload
6.6	FileZilla	
Local/Remote Site	/	
Message Log		
7.		

7.1	9-10	FTP Flow, Passive Port
vsftpd		
7.2	Ubuntu Server LXC, Proxmox Console,	Client
	FileZilla	LAN
7.3	5	
	FileZilla Message Log	
8.		
8.1	Control/Data Connection	Passive Port
8.2		
	vsftpd.conf	FTP
UFW	Transfer Successful	upload
		FileZilla
9.		
9.1	Service Active, Port/Firewall	Login/Directory
	Listing/Upload/Download	Passive Port
9.2		
	Config	FileZilla
	Server	
9.3	FTP Setup Checklist, FileZilla Test Sheet,	Error
10.		
10.1		
10.2		
10.3		
1)		
2)		

	5	5
	31 901-2002	9-10
	FTP Server	2 . 8
Firewall	/ vsftpd, User Permission, Passive Ports, FileZilla	.

1.

Control/Data Connection Passive Mode
vsftpd
Firewall Upload/Download FileZilla

2.

/
2.1 : 40 106
5
1) :
Server
Owner/Permission Firewall
2) : Config Service FileZilla
Client Message Log
Login/Listing/Transfer Error
3) (Performance Evidence):
vsftpd.conf FTP upload UFW
Transfer Successful FileZilla
4) (Knowledge Evidence):
Control/Data Connection Passive Port

2.2 : File
Transfer Service, Permission,
Port Firewall

3.

3.1 FTP Control Data Connection
3.2 vsftpd
3.3 FTP User /

3.4	Passive Port Range	UFW	
3.5	FileZilla	Message Log	
4.			
4.1	FTP	Connection	
4.2	Passive Port	Firewall	
4.3	vsftpd		
4.4	upload	Owner	
4.5	local_enable, write_enable		
chroot_local_user			
4.6	FileZilla		
4.7	Message Log		
5.			
5.1	FTP, Control Connection	Data Connection	
FileZilla		vsftpd	TCP Port 21
Login	USER, PASS	LIST	
Control Connection			Server
Data Connection		Login	
5.2	Passive Mode	Passive Port	
Passive Port		FTP Server	
	Data Connection	Client	40 000-40 100
Server	Control Connection	FileZilla	Client
	Firewall/NAT	Server	Port 21
Passive Port			
5.3	vsftpd		
	local_enable=YES	Linux Login,	
write_enable=YES	chroot_local_user=YES		
	Home Directory		
upload	chown/chmod		
Restart	status		
5.4	FileZilla		
FileZilla	Local Site, Remote Site, Transfer Queue		
Message Log	Login		
Passive Port	Host, Username, Password		
Port 21	Upload/Download		

	Login, Directory Listing, Permission	Data Connection
6.	/	
6.1		Control Connection
Data Connection	FTP	
6.2		Port 21 Login
FileZilla		
6.3	vsftpd	Passive Port 40 000-40 100
	Upload/Download	Message Log
7.		
	Ubuntu Server Guide; vsftpd Manual; FileZilla Client	
Documentation;		
	9-10	
8.	(	)
8.1 Control	Login/	Port 21 Data
8.2	Data Connection	Passive Port Firewall
8.3	Config, UFW,	Server
Successful transfers	FileZilla	

		6
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	Proxmox VE, LXC Container	2 . 13
/ Proxmox Installation, LXC, Production Build, Nginx, PM2 HTTPS		.

1.

Proxmox VE
Server Management Network LXC Container
Web Application Nginx/PM2
HTTPS

2.

/
2.1 : 40 106
5
1) : Proxmox Web GUI
Container IP Web Application
LAN Restart HTTPS Config
2) : Proxmox/Network/Storage/LXC
Production Build, Process, Nginx Configuration
Client
3) (Performance Evidence):
Proxmox Node, LXC Container, Web Application Deploy
Nginx/PM2 Config HTTPS
4) (Knowledge Evidence): Type-1
Hypervisor KVM/LXC Provisioning
Deployment Runbook
2.2 : Virtualization,
Container Platform Deploy Web Application

3.

3.1 Proxmox VE, KVM LXC
3.2 Proxmox Management IP
3.3 Template LXC Container

- 3.4 Deploy Production Build Nginx/PM2
- 3.5 HTTPS/Reverse Proxy LAN

4.

- 4.1 Type-1 Hypervisor KVM/LXC
- 4.2 USB Boot BIOS/UEFI
- 4.3 Hostname, IP, Gateway, DNS Storage
- 4.4 LXC CPU/RAM/Disk/Network
- 4.5 Development Mode Production Build
- 4.6 Nginx/PM2 React Routing 404

4.7 Certificate Redirect HTTP HTTPS

5.

Proxmox VE Type-1 Hypervisor, KVM/LXC, ISO/USB Boot, Management IP, Linux Bridge, Storage, OS Template, LXC Provisioning, React/Node Production Build, Nginx try_files/Reverse Proxy, PM2, SSL/TLS Port 443

6.

- 6.1 Bare-metal Hypervisor
 KVM LXC
- 6.2 USB Boot Proxmox VE
 Hostname, Management IP, Gateway DNS
- 6.3 https://IP:8006 Node/Storage/Bridge
 Template LXC
- 6.4 npm install, npm run build, Nginx Document
Root/try_files PM2 Node.js
- 6.5 Deploy Web Application
LAN Process/Port/Log

6.6 SSL/TLS Self- signed Certificate Port 443 Redirect 80 -> 443 Certificate

7.

- 7.1 11-13 Proxmox VE, LXC
Web Deployment
- 7.2 Server , Proxmox ISO, USB Drive, Switch,
Client Browser Source Code

7.3 6 Installation Checklist, LXC Worksheet

Deployment Runbook

8.

8.1

Provisioning Type-1 Hypervisor KVM/LXC
Deployment Runbook

8.2

Proxmox Node, LXC Container, Web Application Deploy

Nginx/PM2 Config HTTPS

9.

9.1

Application GUI LXC Network
LAN Config

Restart

9.2

Resource/Network/Port/Log Restart/Access
Runbook

9.3

Proxmox Installation Checklist, LXC/Deployment Rubric,
HTTPS Test Sheet

10.

10.1

10.2

10.3

1)

2)

	6	6
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	Proxmox VE, LXC Container	2 . 13
/ Proxmox Installation, LXC, Production Build, Nginx, PM2 HTTPS		.

1.

Proxmox VE
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Web Application Nginx/PM2
HTTPS

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1) : Proxmox Web GUI
Container IP Web Application
LAN Restart HTTPS Config
2) : Proxmox/Network/Storage/LXC
Production Build, Process, Nginx Configuration
Client
3) (Performance Evidence):
Proxmox Node, LXC Container, Web Application Deploy
Nginx/PM2 Config HTTPS
4) (Knowledge Evidence): Type-1
Hypervisor KVM/LXC Provisioning
Deployment Runbook
2.2 : Virtualization,
Container Platform Deploy Web Application

3.

3.1 Proxmox VE, KVM LXC
3.2 Proxmox Management IP
3.3 Template LXC Container

3.4	Deploy Production Build	Nginx/PM2	
3.5	HTTPS/Reverse Proxy		LAN
4.			
4.1	Type-1 Hypervisor		KVM/LXC
4.2	USB Boot	BIOS/UEFI	
4.3	Hostname, IP, Gateway, DNS		Storage
4.4	LXC	CPU/RAM/Disk/Network	
4.5	Development Mode		Production Build
4.6	Nginx/PM2	React Routing 404	
4.7	Certificate	Redirect HTTP	HTTPS
5.			
5.1	Proxmox VE, KVM	LXC	
	Proxmox VE	Type-1 Hypervisor	
		Web GUI KVM	
	Kernel	LXC	Kernel Linux
			Host
5.2		LXC Container	
		ISO/USB	Virtualization
	BIOS	Proxmox	Management IP
	https://IP:8006		Storage, Linux Bridge,
	OS Template	LXC	CPU,
	RAM, Disk, Password/SSH Key	Network	
5.3	Deploy Web Application		
	Development Mode		
	Production Build		
	React	npm run build	Nginx
	Static File	try_files \$uri /index.html	
	Client-side Routing	Node.js	PM2
	Process		Restart
	5.4 HTTPS	Nginx Reverse Proxy	
	HTTPS	TLS	Client/Server
	Certificate	Public Key Self-signed	Lab
	Browser	Nginx	Port 443

Certificate/Private Key
 Redirect Port 80
 HTTPS nginx -t Client
 6. /
 6.1 KVM LXC Kernel
 6.2 LXC Template
 Console Network
 6.3 Deploy Web Application Nginx/PM2 HTTPS
 Runbook Restart

7.

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Proxmox Web GUI	https://IP:8006	Node, Storage, Network LXC
LXC Network	ip a / ip route / ping	IP, Gateway
GitHub Clone/Build	git clone / npm ci / npm run build	
Nginx Config	nginx -t / systemctl reload nginx	Config
Node.js Process	pm2 start / pm2 save / pm2 status	Process
HTTPS	curl -k https://IP / ss -tulpn	Port 443

Proxmox VE Administration Guide; Nginx Documentation; PM2 Documentation; MDN TLS;

11-13

8. ()
 8.1 KVM OS LXC
 Linux Service
 8.2 Template -> Create CT -> Resource -> Network -> Start -> Console -> update -> test IP/DNS
 8.3 Config/Process Runbook

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	Clone GitHub Git Server	2 . 8
/ Git Installation, GitHub Public Repository, HTTPS Clone, Dependencies Deployment		.

1.

Public Repository Git Ubuntu Server HTTPS URL
git clone Dependencies

PM2 Nginx

2.

2.1 : /
5 40 106

1) : Git Clone Repository
README.md

package.json

2) Process, Port, curl Browser
: Checklist

Client

3) (Performance Evidence):
Clone GitHub Dependencies

PM2/Nginx

4) (Knowledge Evidence):
GitHub URL Server

2.2 :

3.

3.1 Public Repository,

HTTPS URL git clone

3.2 Git Ubuntu Server
3.3 Clone GitHub Home Directory Container

3.4 README.md, package.json Dependencies

3.5 Clone PM2 Build Nginx

4.
4.1 Git Server
GitHub Repository

4.2 Repository Public HTTPS URL

4.3 Git apt git --version

4.4 git clone

4.5 README.md package.json

4.6 npm ci npm install

Repository

4.7 Process, Port

5.

Git GitHub ,
Public Repository, HTTPS URL, git clone, RE DME.md,
package.json, package-lock.json, Dependencies, npm ci,
npm install, Node.js pplication, PM2, React/Vite
Production Build, Nginx, Process, Port, curl, Browser
Error

6.

6.1 GitHub

Repository, HTTPS URL Clone

6.2	Repository
GitHub	Proxmox Container
6.3	Public Repository
URL	Proxmox GUI Console
6.4	git clone
	README.md, package.json
6.5	Dependencies
PM2	Build
6.6	Process, Port, curl
	Error
	URL, Network, npm, PM2
Checklist	
7.	
7.1	7
Clone	GitHub
	Server
7.2	Public GitHub Repository, Proxmox/LXC, Ubuntu Server,
	Proxmox GUI Console, Node.js, PM2, Nginx
7.3	7
	Clone
	Checklist
	Repository, Dependencies, Process, Port
8.	
8.1	
	GitHub URL
	Server
8.2	
	Clone
Dependencies	PM2/Nginx
9.	GitHub
9.1	
Clone Repository	
Dependencies	
	Process, Port

9.2

Checklist

Browser

9.3

Git Clone Checklist, Deployment Checklist,

7

10.

10.1

10.2

10.3

1)

2)

	7	7
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	Git Clone Server GitHub	2 . 8 .
/ Git Installation, GitHub Public Repository, HTTPS Clone, Dependencies Deployment		

1.

Public Repository Git Ubuntu Server HTTPS URL
git clone Dependencies

PM2 Nginx

2.

2.1 : 40 106

5

1) : Git Clone Repository
README.md

package.json

Process, Port, curl

Browser

2) : Checklist

Client

3) (Performance Evidence):

Clone GitHub Dependencies

PM2/Nginx

4) (Knowledge Evidence):

GitHub URL Server

2.2 :

3.

- 3.1 Public Repository, HTTPS URL
git clone
- 3.2 Git Ubuntu Server
- 3.3 Clone GitHub Home Directory Container
- 3.4 README.md, package.json Dependencies
- 3.5 Clone PM2 Build Nginx

4.

4.1

- Git**
- Server Repository** **GitHub**
- 4.2 Repository Public
HTTPS URL
- 4.3 Git apt git --version
- 4.4 git clone
- 4.5 README.md package.json
- 4.6 npm ci npm install
Repository
- 4.7 Process, Port

5.

- 5.1 Git, GitHub Repository Clone
Git Ubuntu Server GitHub
- Repository
- Public Repository
Repository Clone HTTPS URL
- Server
- 5.2 Git HTTPS URL
Git sudo apt update sudo apt install git -y
git --version Proxmox GUI Console

Repository Code HTTPS URL .git
 Home Directory git clone URL
 Cloning into

Repository

5.3

Clone

Dependencies

ls -la

README.md

package.json

scripts

node_modules

Repository

npm ci

package-lock.json

npm install

lock file

5.4

Node.js Application

start script

PM2

Process

pm2 status, ss

curl

React/Vite Static Website

npm run build

dist

/var/www/html

nginx -t

reload Nginx

URL/Network, Clone, Dependencies, Process, Port

Nginx

6.

/

6.1

Public Repository

GitHub

Ubuntu Server

6.2

Git, Clone Repository,

README/package.json

Dependencies

6.3

Clone

7.

GitHub Docs: Cloning a repository; Git Documentation:

git-clone; npm Docs: npm ci; PM2 Documentation; Nginx

Documentation;

7

8.

(

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8.1

Repository ->

HTTPS URL ->

Git -> git clone URL ->

8.2

: sudo apt update, sudo apt install git -y, git --
 version, cd ~, git clone URL, cd REPOSITORY, ls -la, cat

README.md, cat package.json

npm ci

npm install

8.3

README/package.json, npm

error, pm2 status/logs, ss -lntp, curl localhost

Nginx

config/port

